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ATHARVA COLLEGE OF ENGINEERING
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Department of Computer Engineering

BE-PROJECT-I LOG BOOK

FOOTBALL VISION AI: COMPUTER VISION AND IOT POWERED AUTONOMOUS VIDEO ASSISTANT REFEREE (VAR)

Academic Year: 2025 – 2026 | Semester VII

GROUP MEMBERS

1.	**Mann Sutariya**	(Roll No: 47)
2.	**Tejas Walanj**	(Roll No: 55)
3.	**Karan Sonar**	(Roll No: 44)
4.	**Yashwin Rajpurohit**	(Roll No: 09)

Varsha Patil

SUPERVISOR / GUIDE

PROF. VARSHA PATIL

Assistant Professor, Department of Computer Engineering

INSTITUTE VISION & MISSION

Vision

To develop a sustainable ecosystem of technical education by fostering innovation, advancing research and leveraging technology with ethical and social values.

Mission

1. To transform the traditional approach of engineering education by experiential learning and interdisciplinary research for professional development.
2. To promote, encourage and facilitate the use of emerging technology and its application to bridge the gap between industry and academia.
3. To imbibe core values of excellence, integrity, teamwork, professional ethics and societal responsibilities.

DEPARTMENT OF COMPUTER ENGINEERING

Vision

To enhance the standard of Computer Engineering graduates with strong academics, research and innovative technology to converge the emerging needs of the industry and society along with ethical values.

Mission

1. To create a productive teaching and learning atmosphere that will foster research initiatives, inventions by using modern tools.
2. To develop and channelize the creative talents of graduates to achieve the evolving needs of the software industry.
3. To inculcate problem solving, team building skills and promote life-long learning with a sense of imparting human values.

PROGRAM OUTCOMES PO'S

PO Code	PO Title	Statement
PO1 **EK**	**Engineering knowledge**	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2 **PA**	**Problem analysis**	Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3 **DS**	**Development/ Design of solutions**	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4 **CP**	**Conduct investigations of complex problems**	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5 **MT**	**Modern tool usage**	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6 **EG**	**The engineer and society**	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7 **ES**	**Environment and sustainability**	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8 **ET**	**Ethics**	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9 **IT**	**Individual and team work**	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10 **CO**	**Communication**	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11 PM	**Project management and finance**	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12 LL	**Life-long learning**	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs):

- **PSO1:** Improving professional and problem-solving abilities to use cutting-edge technologies for solving global issues.
- **PSO2:** Graduates will be able to provide novel solutions by incorporating ethical behavior, effective communication skills and an entrepreneurial spirit.

Program Educational Objectives (PEOs):

- **PEO1:** To prepare competent graduates having strong fundamentals, domain knowledge with modern technology to provide effective solutions for engineering problems.
- **PEO2:** Graduates will have successful technical and professional careers in industry, academia, government and entrepreneurship.
- **PEO3:** To prepare committed and motivated graduates with an investigative approach for lifelong learning.

COURSE OUTCOMES

CO1. To develop the understanding of the problem domain through extensive review of literature.

CO2. To Identify and analyze the problem in detail to define its scope with problem specific data.

CO3. To know various techniques to be implemented for the selected problem and related technical skills through feasibility analysis.

CO4. To design solutions for real-time problems that will positively impact society and the environment.

CO5. To develop clarity of presentation based on communication, teamwork and leadership skills.

CO6. To inculcate professional and ethical behavior.

CO-PO-PSO MAPPING

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	✓			✓									✓	✓
CO2	✓	✓		✓	✓								✓	✓
CO3		✓	✓		✓								✓	✓
CO4	✓	✓		✓		✓	✓						✓	✓
CO5											✓		✓	✓
CO6								✓	✓	✓				✓

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STUDENT INFORMATION

PROJECT TITLE: Football Vision AI – Computer Vision and IoT Powered Autonomous Video Assistant Referee (VAR)

Category of Project: Inhouse / Societal / Research based

Application Domain: Sports Analytics / Autonomous Video Refereeing / Smart Stadiums

Technical Domain: Computer Vision (OpenCV, MOG2, YOLOv8, Homography/Perspective Transformation), IoT & Wearable Biometrics (FastAPI WebSocket streaming, ESP32 telemetry), Full-Stack Web Development (FastAPI, React 19, Tailwind CSS)

FIELD	STUDENT 1	STUDENT 2	STUDENT 3	STUDENT 4
STUDENT ID / EMAIL	mannsutariya-cmpn@atharvacoe.ac.in	tejaswalanj-cmpn@atharvacoe.ac.in	karansonar-cmpn@atharvacoe.ac.in	yashwinrajpurohit-cmpn@atharvacoe.ac.in
NAME	**Mann Sutariya**	**Tejas Walanj**	**Karan Sonar**	**Yashwin Rajpurohit**
ROLL NO & CLASS	47 (BE-CMPN-2)	55 (BE-CMPN-2)	44 (BE-CMPN-2)	09 (BE-CMPN-2)
CONTACT NO.	+91 98201 44521	+91 91374 88293	+91 98923 66104	+91 93245 77189
ADDRESS	B-302, Gokul Dham Heights, Mahavir Nagar, Kandivali West, Mumbai, 400067	12/A, Shanti Niketan CHS, SV Road, Malad West, Mumbai, 400064	C-104, Sai Heritage, Link Road, Borivali West, Mumbai, 400092	Flat 401, Shree Ganesh Tower, Evershine Nagar, Malad West, Mumbai, 400064

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INSTRUCTIONS TO STUDENTS:

1. The logbook must be submitted to the Guide or Co-Guide for verification and evaluation of project activities at least once in a week.
2. Log book duly signed by a guide must be submitted with a project report for evaluation at the end of semester to the department.

DECLARATION

I declare that this project represents my ideas in my own words without plagiarism and wherever others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my project work. I promise to maintain minimum 75% attendance, as per the University of Mumbai norms. I understand that any violation of the above will be cause for disciplinary action by the Institute.

Yours Faithfully,

1. Mann Sutariya	Mann S.
2. Tejas Walanj	Tejas W.
3. Karan Sonar	Karan S.
4. Yashwin Rajpurohit	Yashwin R.

(Date & Signature of Students)

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LETTER OF ACCEPTANCE

I undersigned, **Prof. Varsha Patil** working in Department of Computer Engineering, willing to guide the project titled **"Football Vision AI: Computer Vision and IoT Powered Autonomous Video Assistant Referee (VAR)"** for the BE-Project-I Semester VII for the Academic Year 2025-26. The names of the students are:

1. **Mann Sutariya** (Roll No: 47, BE-CMPN-2)
2. **Tejas Walanj** (Roll No: 55, BE-CMPN-2)
3. **Karan Sonar** (Roll No: 44, BE-CMPN-2)
4. **Yashwin Rajpurohit** (Roll No: 09, BE-CMPN-2)

Varsha Patil	*Varsha Patil*	*M. Patil*
Prof. Varsha Patil **(Project Guide)**	**Prof. Varsha Patil / Dr. Suvarna P.** **(BE-Project I Coordinator)**	**Prof. Mahendra Patil** **(HOD-CMPN)**

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ACE/CMPN/FR/16/2025-26

DATE: 17/07/2025

SCHEDULE FOR BE PROJECT-I

Date	Week	Contents	Remark	Guide Sign
17/07/25	**1**	Basic Concept Presented & Literature Brainstorming	**Completed**	VP
24/07/25	**2**	Title Approved: Autonomous VAR Using Computer Vision	**Done**	VP
31/07/25	**3**	Review of Research Papers & Video Tracking Methodologies	**Working**	VP
07/08/25	**4**	Finalizing Project Scope, Objectives & Feasibility Study	**Done**	VP
14/08/25	**5**	Literature Survey Formulation & Evaluation of FIFA Standards	**Working**	VP
21/08/25	**6**	Study of Existing VAR Systems, Research Gap & Hardware Plan	**Done**	VP
28/08/25	**7**	Proposed System Architecture & Block Diagram Design	**Working**	VP
04/09/25	**8**	Algorithm & Process Design (Detection & Homography Mapping)	**OK**	VP
11/09/25	**9**	Core Vision Pipeline Implementation & Ball/Player Tracking	**OK**	VP
18/09/25	**10**	FastAPI Backend Streaming & Event Triggering Development	**Done**	VP
25/09/25	**11**	React Dashboard Integration & Video Control Room UI	**Working**	VP
02/10/25	**12**	Interim Project Report & Documentation Formulation	**Done**	VP
09/10/25	**13**	End-to-End System Testing, Latency Optimization & Adjustments	**Done**	VP
16/10/25	**14**	Final Review Presentation, Working Demo & Logbook Submission	**Done**	VP

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ACE/CMPN/FR/16/2025-26
DATE: 17/07/2025

PROGRESS/ATTENDANCE REPORT

Group No.: 03
Title of the Project: Football Vision AI: Computer Vision and IoT Powered Autonomous Video Assistant Referee (VAR)
Name of Student 1: Mann Sutariya | **Name of Student 2:** Tejas Walanj
Name of Student 3: Karan Sonar | **Name of Student 4:** Yashwin Rajpurohit
Name of the Supervisor/Guide: Prof. Varsha Patil

Sr. No	Date	Attendance				Progress / Suggestion	Mapping		
		1	2	3	4		CO	PO	PSO
1	17/07/25	P	P	P	P	Working on project title finalization, domain selection, and scope identification for AI-powered VAR system.	CO1	PO1 PO6	PSO1
2	24/07/25	P	P	P	P	Conducted comprehensive literature survey on computer vision in sports officiating, Hawk-Eye protocols, and deep neural models.	CO1 CO2	PO2 PO4	PSO1 PSO2
3	31/07/25	P	P	P	P	Study and acquisition of broadcast football video datasets (SoccerNet clips) and camera calibration feeds.	CO2 CO3	PO3 PO5	PSO1

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Sr. No	Date	Attendance				Progress / Suggestion	Mapping		
		1	2	3	4		CO	PO	PSO
4	07/08/25	**P**	**P**	**P**	**P**	Detailed requirement analysis (hardware specs, video capture parameters, and IoT alert triggers).	**CO2** **CO4**	PO1 PO3 PO5	PSO1 PSO2
5	14/08/25	**P**	**P**	**P**	**P**	Deployment of OpenCV background subtractor (MOG2) and object detection pipeline for players and ball.	**CO3** **CO4**	PO2 PO3 PO5	PSO1
6	21/08/25	**P**	**P**	**P**	**P**	Multi-player tracking, centroid association, and team jersey color thresholding in HSV color space.	**CO3** **CO4**	PO3 PO5	PSO1
7	28/08/25	**P**	**P**	**P**	**P**	Pitch keypoint detection and perspective transformation (homography matrix to 2D top-down pitch view).	**CO3** **CO4**	PO2 PO3 PO5	PSO1 PSO2
8	04/09/25	**P**	**P**	**P**	**P**	Automated offside line projection and goal-line boundary crossing evaluation algorithms.	**CO4**	PO3 PO5	PSO1
9	11/09/25	**P**	**P**	**P**	**P**	FastAPI streaming backend development with MJPEG video feed endpoints and event logging database.	**CO4**	PO3 PO5	PSO1
10	18/09/25	**P**	**P**	**P**	**P**	Design and implementation of React.js VAR Studio dashboard with multi-camera switching and decision alerts.	**CO4** **CO5**	PO5 PO10	PSO1 PSO2
11	25/09/25	**P**	**P**	**P**	**P**	System benchmarking, latency evaluation under broadcast feeds, and IoT alert mechanism testing.	**CO4** **CO6**	PO8 PO11	PSO1 PSO2
12	09/10/25	**P**	**P**	**P**	**P**	Created complete logbook, completed BE-Project-I report documentation, and conducted presentation dry-run.	**CO5** **CO6**	PO10 PO11 PO12	PSO1 PSO2

Varsha Patil

Name, Date & Sign of the Supervisor/Guide

(Prof. Varsha Patil)

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REVIEW-I FORM

Group No.: 03

Title of BE-Project I: Football Vision AI: Computer Vision and IoT Powered Autonomous Video Assistant Referee (VAR)

Date of Review-I: 22/08/2025

No. of students in project team: 4

Student BE-Project I Performance Analysis (Put Tick as per your Observation)

Sr. No.	Observation	Excellent (3)	Very Good (2)	Good (1)
		(3)	(2)	(1)
1	Quality of problem and Clarity	✓		
2	Clarity of Problem definition based on need.	✓		
3	Innovativeness in solutions	✓		
4	Feasibility of proposed problem solutions and selection of best solution		✓	
5	Cost effectiveness and Societal impact	✓		
6	Overall Presentation & Performance	✓		

Comments:

Working on papers and authentication paper. More papers to be surveyed on sports video analytics and pitch homography algorithms. Proposed architecture is sound and approved.

Project Guide & Panel Members Signature:

1) Prof. Varsha Patil 2) Prof. Pranali B. 3) Prof. Anuja G.

Varsha Patil

Prof. Varsha Patil / Dr. Suvarna P.
Project Coordinator

M. Patil

Prof. Mahendra Patil
HOD-CMPN

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REVIEW-II FORM

Group No.: 03

Title of BE-Project I: Football Vision AI: Computer Vision and IoT Powered Autonomous Video Assistant Referee (VAR)

Date of Review-II: 03/10/2025

No. of students in project team: 4

Student BE-Project I Performance Analysis (Put Tick as per your Observation)

Sr. No.	Observation	Excellent (3)	Very Good (2)	Good (1)
		(3)	(2)	(1)
1	Usage of effective skill sets	✓		
2	Effective use of standard engineering norms	✓		
3	Contribution of an individual's as member or leader		✓	
4	Clarity in written and oral communication	✓		
5	Functioning of working model as per stated requirements	✓		
6	Overall Presentation & Performance	✓		

Comments:

Working on documentation and interfacing of back-end (FastAPI streaming engine) with front-end (React VAR Dashboard). Working model demonstrated effectively. Ready for final examination.

Project Guide & Panel Members Signature:

1) Prof. Varsha Patil 2) Prof. Pranali B. 3) Prof. Anuja G.

Varsha Patil
Prof. Varsha Patil / Dr. Suvarna P.
Project Coordinator

M. Patil
Prof. Mahendra Patil
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EXAMINER'S FEEDBACK FORM

Name of External examiner: Dr. Nazneen Ansari
College of External examiner: SFIT, Borivali
Name of Internal examiner: Prof. Varsha Patil
Date of Examination: 27/10/2025
No. of students in project team: 4

Student Performance Analysis (Put Tick as per your Observation)

Sr. No.	Observation	Excellent (3)	Very Good (2)	Good (1)
		(3)	(2)	(1)
1	Quality of problem and Clarity	✓		
2	Clarity of Problem definition based on need.	✓		
3	Innovativeness in solutions		✓	
4	Feasibility of proposed problem solutions and selection of best solution	✓		
5	Cost effectiveness and Societal impact	✓		
6	Usage of effective skill sets		✓	
7	Effective use of standard engineering norms		✓	
8	Contribution of an individual's as member or leader	✓		
9	Clarity in written and oral communication	✓		
10	Full functioning of working model as per stated requirements	✓		
11	Overall Presentation & Performance	✓		

Dr. Nazneen A.
Dr. Nazneen Ansari
External Examiner
27/10/2025

Varsha Patil
Prof. Varsha Patil
Internal Examiner
27/10/2025

M. Patil
Prof. Mahendra Patil
HOD-CMPN